Dequantization on Geometry

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I'm interested in the re-interpretation of non-linear PDE from algebraic geometry and QFT perspective. It's typically Navier-Stokes, Apery, period integral of Landau Ginzburg, Monge-Ampere below, and I consider possible strategies are 1: I consider non-linear PDE differential geometrically, by making some connection form diffyq $\nabla f = 0$ on a real smooth manifold X, possibly symplectification (X', ω) if necessary, where I further put geometric quantiztion, and 2: Derived moduli space of solutions spaces of non-linear PDE, whose alegbraic geometric realization, and 3: Transform non-linear PDE to CFT, corresponding to algebraic geometry.

- Navier Stokes

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}$$

  Or

$$_{\mu}T^{\mu\nu} = 0$$

- Apery (Quantum Satake)

$$\mathcal{L}_{\mathrm{Ap}} = \theta^3 - z(2\theta+1)(17\theta^2+17\theta+5) + z^2(\theta+1)^3, \qquad \theta = z\frac{d}{dz}.$$

- Landau-Ginzburg (Period Integral, SCFT)

$$\int e^{-f(x)} \qquad f(x) = \text{polynomial}$$

- Monge-Ampere (SYZ)

$$\det\left(\frac{\partial^2 \phi}{\partial x_i \partial x_j}\right) = f(x)$$

- SUGRA (String Theory)

$$R_{\mu\nu} - \frac{1}{2}R\, g_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

Remark 0.1

I also noticed several different approaches, Kcryzka introduces derived algebraic geometry realization of jet spaces, Ladychenskaya formula for solving 2d NSE. These are quite interesting but quite different from my study.

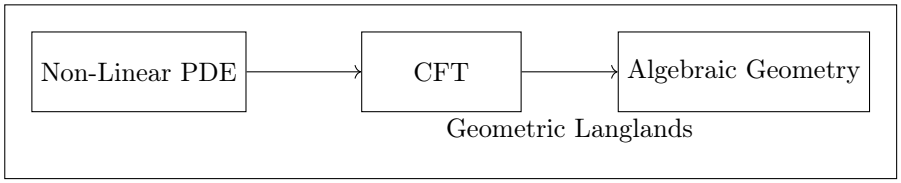
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1 Navier-Stokes



Proposition 1.1. NSE-CFT Conversion

TBA

Definition 1.1. Quantization of NSE

2 Apery

Definition 2.1. Geometric Satake

$$\mathrm{Perv}(\mathrm{Gr}_G) \simeq \mathrm{Rep}(G^\vee).$$

Definition 2.2. Quantum Satake

Definition 2.3. 4D/2D CFT Correspondence

Definition 2.4

Definition 2.5

Definition 2.6

3 Quantization

Definition 3.1. Prequantization

Prequantization is a process of making an operator $Q(f)$ from a smooth function f, $Q : C^\infty(X) \to End(\mathcal{H})$ on a symplectic manifold (X, ω). More precisely, $\mathcal{H} = \Gamma^\infty(L)$ where L is a prequantum line bundle with a connection ∇_{X_f}, and an operator given by $Q(f) := -i\hbar\nabla_{X_f} + f$, so that bracket satisfies $[Q(f), Q(g)] = i\hbar Q(\{f, g\})$.

Typically, the operator is Kostant–Souriau operator $Q(f) := -i\hbar(X_f + \frac{1}{i\hbar}\theta(X_f)) + f$.

Definition 3.2

Definition 3.3

References

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